Fiscal Year:	FY 2016	**Task Last Updated:**	FY 03/30/2016
PI Name:	Datta, Kamal M.D.		
Project Title:	High-LET Radiation, Reproductive Hormones and Chronic Inflammation: An Integrated Approach to Carcinogenesis Risk Estimate		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20057-0001	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	01/01/2013	**End Date:**	12/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2015, per NSSC information (Ed., 12/17/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	Colorectal cancer (CRC) is the third most common cancer in the USA accounting for 10% of all cancer deaths. In 2016, the American Cancer Society estimated that 95,270 new cases of CRC will be diagnosed, with 39,220 deaths. Approximately 60% of new cases will be in males and the frequency of precancerous lesions being present in about 10% of adults at age 40 (younger than the average age of a typical astronaut). Epidemiological studies in atomic bomb survivors and in radiation workers implicate radiation as a risk factor for CRC. Due to the high frequency of CRC in the U.S. and high linear energy transfer (high-LET) characteristics of heavy ion radiation, an even small increase by space travel will have a major impact on risk estimates and planning of future long-term missions into space. Importantly, unlike low-LET radiation, CRC risk prediction of high-LET radiation is problematic due to the paucity of in vivo long-term follow up data in human or animal models. In our CRC mouse model studies we have already reported that exposure to high-LET 56-Fe radiation, relative to low-LET exposure, resulted in higher intestinal tumorigenesis in APCMin/+ and APC1638N/+ mice. Although epithelial cells are the origin of gastrointestinal (GI) cancer, we are yet to clearly understand how high-LET heavy ion radiation differentially altered GI epithelial cell homeostasis relative to low-LET radiation to promote greater tumorigenesis. Therefore, acquiring in vivo data will help in developing risk models of high-LET radiation exposure not only to lessen the uncertainty of health risks of space travel but also to develop strategies to protect healthy tissues from other sources of high-LET radiation such as environmental radon and radiotherapeutic heavy ion radiation exposures. The overall goal of this proposal is to understand the effects of cosmic radiation on persistent oxidative stress and inflammation in the GI tract and how these alterations relate to GI carcinogenesis. In conjunction with our GI NASA Specialized Center of Research (NSCOR) program, the current proposal aims to compare the results of highly energetic charged (HZE) particle radiation to gamma radiation in mice to improve NASA's carcinogenesis risk estimates for GI cancer in astronauts through a mechanistic understanding of the underlying biological processes. The specific aim for the proposal is therefore to characterize the roles of HZE radiation-induced persistent oxidative stress and inflammatory responses in gastrointestinal carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S., and American Cancer Society estimates that there will be 95,270 new cases of colorectal cancer (CRC) with 39,220 persons predicted to die of the disease in 2016. Colorectal cancer when considered in men and women is the second leading cancer killer in the United States (http://www.cdc.gov/). Considering the high frequency of colorectal tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. Inflammation and cellular stress are critical elements of carcinogenesis and both are associated with alterations in biomolecules such as DNA. Because radiation is also known to initiate cellular oxidative stress and inflammatory responses, HZE radiation due to its qualitative difference from terrestrial gamma radiation is expected to induce long-term stress and inflammatory responses, thus increasing the risk of GI cancer.
Task Progress:	Inflammation and oxidative stress is intimately linked and both are strongly implicated in gastrointestinal (GI) carcinogenesis, and exposure to radiation results in production of reactive oxygen species (ROS) leading to oxidative stress. High levels of ROS are known to induce an apoptotic response but sustained sub-lethal oxidative stress has the potential to produce persistent modification of biomolecules including proteins, DNA, and lipids leading to alterations in their functions with adverse cellular consequences. Delayed effects of radiation exposure have been attributed to persistent oxidative stress in tissues as a result of functional changes at the molecular level. Our study was focused on 1. How HZE radiation affects cellular redox balance and inflammation at the tissue level in the GI tract, and how do these effects compare to spontaneous and gamma-ray mediated alterations? 2. What is the relative importance of radiation quality on the extent and nature of persistent cellular changes, with a special emphasis on tissue inflammation and cellular mitochondrial function? 3. How do these changes relate to risk of GI tumorigenesis and can an understanding of these processes inform strategies to reduce risk? Specifically, we investigated molecular mechanisms that could contribute to carcinogenic risk to the human GI tract. The overall aim of this project was to characterize the effects of heavy ion space radiation in triggering persistent stress and inflammatory responses with implications for GI tumorigenesis. We focused on quantitative and qualitative measurements of long-term cellular stress responses, especially oxidative, metabolic, inflammatory, and oncogenic stress responses in serum and in GI tissues after 56-Fe- and gamma-radiation. We also worked on effects of HZE radiation exposure on intestinal stem cell in relation to cellular stress. Tasks completed are - i] Persistent oxidative stress in intestinal epithelial cells (IEC), ii] Downregulation of autophagy, iii] Upregulation of leptin/IGF1 signaling axis, iv] Standardized intestinal stem cell (ISC) isolation protocol, v] Measure oxidative stress in ISC, vi] Assess proliferative signal in ISC, vii] Investigate effects of heavy ion radiation on intestinal epithelial cell (IEC) migration. Considering that this project is integrated with the ongoing GI NSCOR, the experiments were aligned to support carcinogenesis studies of the NSCOR.
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